

Gregory Parent

Professor Kirk

ARKEO 2812: Hieroglyphs to HTML: History of Writing

May 18, 2026

AstraGlyph: A Sci-Fi Hybrid Writing System and [Translator](#)

For my final project, I designed AstraGlyph, a science-fiction-inspired hybrid writing system that combines logograms, phonetic signs, and visual design rules. I chose this project because I have always been interested in science fiction movies and books, such as *Dune*, *Arrival*, and the *Star Wars* franchise. The writing systems present in these stories reflect their societies' unique ways of thinking and culture, an idea I was interested in and incorporated into AstraGlyph. Additionally, rather than simply making a new alphabet where each English letter has a replacement symbol, I wanted to create a script that felt as if it had evolved from modern English into something more visual, symbolic, and futuristic. Drawing from my background as a Computer Science major, I also wanted the project to have an interactive digital form, so I built a [website translator](#) that can convert English words and sentences into my writing system.

AstraGlyph is a hybrid writing system as some of its signs are logograms, representing whole ideas or concepts. I decided to make the logograms centered around science-fiction words and concepts such as planets, stars, ships, artificial intelligence, signals, colonies, and danger, as these are concepts that might appear more often in a futuristic space society. The system also includes phonetic characters for names, unfamiliar words, and proper nouns. For example, if a word does not have its own logogram, the translator breaks it into phonetic glyphs using vowels, consonants, and blended sounds like “SH,” “TH,” “TR,” and “QU.” This makes AstraGlyph more flexible than a purely symbolic system, because it can represent both meaning and sound.

When designing the characters, I decided to make the visual design based on simple geometric motifs. Circles represent worlds, systems, bodies, and cycles. Ellipses suggest orbits, gates, and transmissions. Squares are connected to machines, code, storage, and artificial intelligence. Triangles represent direction, movement, ships, action, and warning. Dots suggest stars, nodes, settlements, or individual units, while lines show motion, connection, and communication. The idea behind this was that if someone was unfamiliar with a specific character but understood the essence of the system, they would potentially be able to deduce what the unknown logogram meant. Additionally, I intended to keep the symbols simple enough such that they could plausibly function as a writing system, but distinct enough that they would feel like they belonged to a science-fiction world.

I had two major sources that influenced my project, the first being Denis Villeneuve's 2016 film, *Arrival*, which inspired me to think about writing as something more than a linear recording of speech. The heptapod writing in the film is visually striking because it challenges the normal left-to-right structure of English and connects writing to perception, time, and alien thought. While AstraGlyph is not circular in the same exact way, *Arrival* influenced my goal of making the script feel like it came from a culture with different priorities than our own. The second influence was Charles K. Bliss's Blissymbolics, a constructed system of symbols meant to communicate concepts visually rather than through one spoken language. Blissymbolics helped me think about how abstract ideas can be represented through simplified visual signs. It influenced my use and creation of the logograms and my attempt to make the symbols understandable through recurring visual patterns instead of random decoration.

This project relates closely to the course because it explores many of the same questions that appear throughout the history of writing systems. For example, by choosing to make

AstraGlyph a mix of logograms and phonetic characters, I explore how writing systems do not all work the same way. Some writing systems primarily represent sounds, while others represent syllables, morphemes, words, or broader concepts. AstraGlyph reflects this history by combining phonetic signs with logograms, similar to how many historical writing systems, such as Japanese, are not purely one type. It also connects to the end of the course and our discussions of typewriters, fonts, emoji, and digital writing. My project begins with the ancient idea of meaningful visual signs, but presents those signs through a modern computer interface.

The main point my project makes is that writing is not just a tool to record speech. A writing system reflects the needs, values, and worldview of the culture that creates it. For a space exploring civilization, it would make sense for writing to emphasize concepts like planets, ships, signals, artificial intelligence, and danger. A civilization communicating across planets, species, and languages might not rely only on an alphabet, as it might need a system that can communicate meaning visually while still having a phonetic backup for names and unknown words. AstraGlyph, therefore, imagines writing as a technology of translation between different kinds of minds. The digital form of the project is important because the translator shows how the writing system works in practice. When a user types an English sentence, the program decides whether each word should become a logogram or be broken down phonetically. Building the project as a computer program also reinforces the futuristic premise of AstraGlyph, since the writing system exists through a digital interface rather than only on paper.

The process of creating AstraGlyph involved several stages. First, I developed the basic concept of a sci-fi hybrid script. Then I created a list of logograms, organized into categories such as space, technology, life, and abstract concepts. After that, I designed the phonetic fallback system with vowels, consonants, and blended sounds. One exciting moment was realizing that

the script did not need to be a complete language to work as a writing system. By focusing on the characters rather than inventing every part of grammar and speech, I was able to make the project more focused and realistic. One challenge was balancing creativity with consistency. If every glyph looked completely different, the system would feel random. However, if every symbol were too simple or obvious, it would feel more like a set of icons than a writing system. I addressed this by creating visual rules that repeat across the system. Another challenge was when programming the translator. I needed to decide how to handle English letters or sounds that do not have direct equivalents, such as C, J, Q, and X. I solved this by treating the system as phonetic rather than alphabetic: C can become K or S depending on sound, J can become D-CH, Q can become QU, and X can become K-S or Z. I decided which substitution to make based on a set list of rules depending on preceding and following letters. This helped make the system feel less like a character-swapping cipher and more like an actual writing system.

Overall, AstraGlyph is my attempt to combine the history of writing systems with science fiction and computer science. It draws on our course's themes of logograms, phonetic signs, visual abstraction, and digital writing, while also imagining how writing might evolve in a future world. The project argues that writing systems are not just a tool, but organize meaning, shape interpretation, and reveal what a culture considers important enough to encode.

Works Cited

Arrival. Directed by Denis Villeneuve, screenplay by Eric Heisserer, performances by Amy Adams, Jeremy Renner, and Forest Whitaker, Paramount Pictures, 2016.

Bliss, Charles K. *Semantography (Blissymbolics): A Simple System of 100 Logical Pictorial Symbols, Which Can Be Operated and Read Like $1+2=3$ in All Languages*. 2nd enlarged ed., Semantography (Blissymbolics) Publications, 1965.